

Explicit Formulae For Generalized Fibonacci Sequences

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February, 2023

Define a ‘generalized Fibonacci sequence’ of order p recursively by adding the preceding p number of terms to get the successor term: (The sequence begins with $p - 1$ zeroes and then a 1)

$$F_n^p = \sum_{k=1}^p F_{n-k}^p \quad (1)$$

We will first construct a generating function for the p th-order Fibonacci sequence and then utilize Laurent Decomposition to extract the coefficients. Thus, let;

$$M_p(x) = \sum_{n=0}^{\infty} F_n^p x^n \Rightarrow x^k M_p(x) = \sum_{n=k}^{\infty} F_{n-k}^p x^n \quad (2)$$

Using;

$$x^k M_p(x) = \sum_{n=p}^{\infty} F_{n-k}^p x^n + \sum_{n=k}^{p-1} F_{n-k}^p x^n$$

One is then motivated to sum over all values of $k \leq p$ to use the defining recurrence relation (1) and this produces;

$$\sum_{k=1}^p x^k M_p(x) = \sum_{k=1}^p \sum_{n=p}^{\infty} F_{n-k}^p x^n + \sum_{k=1}^p \sum_{n=k}^{p-1} F_{n-k}^p x^n \Rightarrow M_p(x) \left[\sum_{k=1}^p x^k \right] = \sum_{n=p}^{\infty} \left[\sum_{k=1}^p F_{n-k}^p \right] x^n + \sum_{k=1}^p \sum_{n=k}^{p-1} F_{n-k}^p x^n$$

In the leftmost sum, all Fibonacci coefficients are of index less than $p - 1$ and as such are immediately 0 by definition.

$$M_p(x) \left(x \frac{1-x^p}{1-x} \right) = \left[\sum_{n=p}^{\infty} F_n^p x^n + \sum_{p=0}^{p-1} F_n^p x^n \right] - \sum_{p=0}^{p-1} F_n^p x^n \Rightarrow M_p(x) \left(x \frac{1-x^p}{1-x} - 1 \right) = - \sum_{p=0}^{p-1} F_n^p x^n$$

$$M_p(x) = \frac{-x^{p-1}}{x \frac{1-x^p}{1-x} - 1} = \frac{x^{p-1} - x^p}{x^{p+1} - 2x + 1} \quad (3)$$

By the standard Laurent Decomposition formula this may be reversed to obtain;

$$F_n^p = \frac{1}{2\pi i} \oint \frac{M_p(z)}{z^{n+1}} dz = \frac{1}{2\pi i} \oint \frac{z^{p-n-2} - z^{(p-n-1)}}{z^{p+1} - 2z + 1} dz$$

This contour will be manifested as a simple circle around the origin with radius r , being careful to keep the circle smaller than the set of zeroes of the denominator. When $p = 2$, the denominator can be solved to evaluate the integrand via the Residue Theorem, however the general case is significantly more messy to evaluate. As such, we proceed via substitution $z = re^{i\theta}$;

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} r e^{i\theta} r^{(p-n-2)} \frac{(e^{i\theta})^{(p-n-2)} - r (e^{i\theta})^{(p-n-1)}}{(r e^{i\theta})^{p+1} - 2(r e^{i\theta}) + 1} d\theta \quad (4)$$

To simplify further, one must first expand and take the real part of both sides:

$$= \frac{r^{(p-n-1)}}{2\pi} \int_{-\pi}^{\pi} (\cos \theta + i \sin \theta) \frac{\cos(p-n-2)\theta - r \cos(p-n-1)\theta - r i \sin(p-n-1)\theta + i \sin(p-n-2)\theta}{r^{(p+1)} \cos(p+1)\theta + r^{(p+1)} i \sin(p+1)\theta - 2r \cos \theta - 2r i \sin \theta + 1} d\theta$$

Grouping terms under shorthand functions;

$$= \Re \left[\frac{r^{(p-n-1)}}{2\pi} \int_{-\pi}^{\pi} \frac{[A(\theta) + iB(\theta)] [\cos + i \sin]}{C(\theta) + iD(\theta)} d\theta \right]$$

$$\frac{r^{(p-n-1)}}{2\pi} \int_{-\pi}^{\pi} \frac{[A(\theta)D(\theta) - B(\theta)C(\theta)] \sin(\theta) + [A(\theta)C(\theta) + B(\theta)D(\theta)] \cos(\theta)}{C(\theta)^2 + D(\theta)^2} d\theta$$

A significant deal of trigonometric combination (angle sum identities primarily) and simplification later, two integral characterisations emerge. The first being:

$$\int_0^{\pi} \frac{r^{(p+1)} (\cos((n+2)\theta) - r \cos((n+1)\theta)) - r (\cos((p-n)\theta) + 2 \cos(\theta(p-n-2))) + (1+2r^2) \cos((p-n-1)\theta)}{\eta(\theta)} d\theta \quad (5)$$

The second, arguably nicer, formula, which is valid only for $(n, p) \geq 1$, comes from first factoring the complex exponential in (4) *before* taking the real part of the expression. The process from that point is nearly identical however. $F_n^P =$

$$\int_{-\pi}^{\pi} \frac{r^{(p+1)} \cos(3\theta) - r (2 \cos((p-3)\theta) + \cos((p-1)\theta)) + (1+2r^2) \cos((p-2)\theta) - r^{(p+2)} \cos(2\theta)}{\eta(\theta)} \cos((n-1)\theta) d\theta \quad (6)$$

Where;

$$\eta(\theta) = \pi r^{(n+1-p)} \left(r^{2(p+1)} + 1 + 4r^2 - 4r \cos \theta + 2r^{(p+1)} \cos((p+1)\theta) - 4r^{(p+2)} \cos(p\theta) \right)$$

Equation (6) is far more neat, workable, and furthermore contains n only in a single, coefficient term, highly reminiscent of a Fourier series integral applied to a kernel containing only p . Any sufficiently small r value works in this expression, however $r = 1/2$ simplifies the integrand rather well. This produces: $\forall(n, p) \geq 1$

$$F_n^p = \frac{2^{n-p}}{\pi} \int_{-\pi}^{\pi} \frac{2^{-p} (\cos(3\theta) - \frac{1}{2} \cos(2\theta)) - \cos((p-1)\theta) + 3 \cos((p-2)\theta) - 2 \cos((p-3)\theta)}{2^{-2(p+1)} + 2(1 - \cos \theta) + 2^{-p} (\cos((p+1)\theta) - \cos(p\theta))} \cos((n-1)\theta) d\theta \quad (7)$$

Though one has no justification to extend $n, p \in \mathbb{N}^2$ to $\mathbb{R}^2$ given that the Laurent Decomposition and Generating Function arguments from above apply only to integral values of these variables, expressions (5) & (6) still numerically exist for arbitrary n, p .